

問題 1

α, β がともに鈍角で, $\tan \alpha = -\frac{1}{2}, \cos \beta = -\frac{4}{5}$ のとき, 次の値を求めよ.

解答:

まず $\sin \alpha, \cos \alpha, \sin \beta, \tan \beta$ を求める.

$$1 + \tan^2 \alpha = \frac{1}{\cos^2 \alpha} \text{ より } 1 + \frac{1}{4} = \frac{1}{\cos^2 \alpha}, \cos^2 \alpha = \frac{4}{5}$$

α は鈍角だから $\cos \alpha < 0$.

$$\cos \alpha = -\frac{2}{\sqrt{5}}, \sin \alpha = \tan \alpha \cos \alpha = \left(-\frac{1}{2}\right)\left(-\frac{2}{\sqrt{5}}\right) = \frac{1}{\sqrt{5}}$$

β も鈍角で $\cos \beta = -\frac{4}{5}$ だから $\sin \beta > 0$.

$$\sin \beta = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}, \tan \beta = \frac{3/5}{-4/5} = -\frac{3}{4}$$

$$\begin{aligned} (1) \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ &= \frac{1}{\sqrt{5}} \cdot \left(-\frac{4}{5}\right) + \left(-\frac{2}{\sqrt{5}}\right) \cdot \frac{3}{5} = -\frac{4}{5\sqrt{5}} - \frac{6}{5\sqrt{5}} \\ &= -\frac{10}{5\sqrt{5}} = -\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5} \end{aligned}$$

$$\begin{aligned} (2) \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \left(-\frac{2}{\sqrt{5}}\right)\left(-\frac{4}{5}\right) - \frac{1}{\sqrt{5}} \cdot \frac{3}{5} = \frac{8}{5\sqrt{5}} - \frac{3}{5\sqrt{5}} \\ &= \frac{5}{5\sqrt{5}} = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5} \end{aligned}$$

$$\begin{aligned} (3) \tan(\alpha - \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\ &= \frac{-\frac{1}{2} - \left(-\frac{3}{4}\right)}{1 + \left(-\frac{1}{2}\right)\left(-\frac{3}{4}\right)} = \frac{\frac{1}{4}}{\frac{11}{8}} = \frac{1}{4} \cdot \frac{8}{11} = \frac{2}{11} \end{aligned}$$

問題 2

$\sin \alpha = -\frac{2\sqrt{2}}{3}, \pi < \alpha < \frac{3}{2}\pi$ のとき, $\sin 2\alpha, \cos 2\alpha, \sin \frac{\alpha}{2}, \cos \frac{\alpha}{2}$ の値を求めよ.

解答:

$\pi < \alpha < \frac{3}{2}\pi$ (第 3 象限) だから $\cos \alpha < 0$.

$$\cos^2 \alpha = 1 - \frac{8}{9} = \frac{1}{9}, \cos \alpha = -\frac{1}{3}$$

2 倍角の公式より

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha = 2 \left(-\frac{2\sqrt{2}}{3}\right) \left(-\frac{1}{3}\right) = \frac{4\sqrt{2}}{9}$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha = 1 - 2 \cdot \frac{8}{9} = -\frac{7}{9}$$

半角の公式を使う前に $\frac{\alpha}{2}$ の範囲を調べる.

$$\pi < \alpha < \frac{3}{2}\pi \text{ の各辺を } 2 \text{ で割って } \frac{\pi}{2} < \frac{\alpha}{2} < \frac{3}{4}\pi$$

これは第 2 象限だから $\sin \frac{\alpha}{2} > 0, \cos \frac{\alpha}{2} < 0$.

$$\sin^2 \frac{\alpha}{2} = \frac{1 - \cos \alpha}{2} = \frac{1 + \frac{1}{3}}{2} = \frac{2}{3}$$

$$\therefore \sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}$$

$$\cos^2 \frac{\alpha}{2} = \frac{1 + \cos \alpha}{2} = \frac{1 - \frac{1}{3}}{2} = \frac{1}{3}$$

$$\therefore \cos \frac{\alpha}{2} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$$

問題 3

次の等式を証明せよ.

解答:

$$(1) \frac{\sin(\alpha + \beta)}{\sin(\alpha - \beta)} = \frac{\tan \alpha + \tan \beta}{\tan \alpha - \tan \beta}$$

加法定理で展開し, 分子・分母を $\cos \alpha \cos \beta$ で割る.

$$\begin{aligned} \text{左辺} &= \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\sin \alpha \cos \beta - \cos \alpha \sin \beta} \\ &= \frac{\frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} + \frac{\cos \alpha \sin \beta}{\cos \alpha \cos \beta}}{\frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} - \frac{\cos \alpha \sin \beta}{\cos \alpha \cos \beta}} = \frac{\tan \alpha + \tan \beta}{\tan \alpha - \tan \beta} = \text{右} \end{aligned}$$

辺

□

$$(2) \tan\left(\frac{\pi}{4} + x\right) \tan\left(\frac{\pi}{4} - x\right) = 1$$

$\tan \frac{\pi}{4} = 1$ に注意して正接の加法定理を使う.

$$\tan\left(\frac{\pi}{4} + x\right) = \frac{1 + \tan x}{1 - \tan x}, \quad \tan\left(\frac{\pi}{4} - x\right) = \frac{1 - \tan x}{1 + \tan x}$$

辺々掛けると

$$\text{左辺} = \frac{1 + \tan x}{1 - \tan x} \cdot \frac{1 - \tan x}{1 + \tan x} = 1 = \text{右辺}$$

□

問題 4

次の 3 倍角の公式を証明せよ.

解答:

$3\theta = 2\theta + \theta$ とみて加法定理と 2 倍角の公式を使う.

$$(1) \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

$$\sin 3\theta = \sin(2\theta + \theta) = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta$$

$$= 2 \sin \theta \cos \theta \cdot \cos \theta + (1 - 2 \sin^2 \theta) \sin \theta$$

$$= 2 \sin \theta \cos^2 \theta + \sin \theta - 2 \sin^3 \theta$$

$$\cos^2 \theta = 1 - \sin^2 \theta \text{ を代入して}$$

$$= 2 \sin \theta (1 - \sin^2 \theta) + \sin \theta - 2 \sin^3 \theta$$

$$= 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - 2 \sin^3 \theta$$

$$= 3 \sin \theta - 4 \sin^3 \theta$$

□

$$(2) \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$\cos 3\theta = \cos(2\theta + \theta) = \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$$

$$= (2 \cos^2 \theta - 1) \cos \theta - 2 \sin \theta \cos \theta \cdot \sin \theta$$

$$= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta \sin^2 \theta$$

$$\sin^2 \theta = 1 - \cos^2 \theta \text{ を代入して}$$

$$= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta (1 - \cos^2 \theta)$$

$$= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta + 2 \cos^3 \theta$$

$$= 4 \cos^3 \theta - 3 \cos \theta$$

□ 問題 7

問題 5

積を和・差に直す公式を用いて、次の式を簡単な形にせよ。

解答:

積和の公式

$$\sin A \cos B = \frac{1}{2} \{ \sin(A+B) + \sin(A-B) \}$$

$$\sin A \sin B = \frac{1}{2} \{ \cos(A-B) - \cos(A+B) \}$$

$$(1) \sin \theta \cos 3\theta + \sin \theta \cos 5\theta + \sin \theta \cos 7\theta$$

各項を直すと $(\sin(-x) = -\sin x$ に注意)

$$\sin \theta \cos 3\theta = \frac{1}{2} (\sin 4\theta - \sin 2\theta)$$

$$\sin \theta \cos 5\theta = \frac{1}{2} (\sin 6\theta - \sin 4\theta)$$

$$\sin \theta \cos 7\theta = \frac{1}{2} (\sin 8\theta - \sin 6\theta)$$

$$\text{辺々加えると } \sin 4\theta \text{ と } \sin 6\theta \text{ が打ち消し合っ}$$

$$= \frac{1}{2} (\sin 8\theta - \sin 2\theta)$$

$$(2) \sin \theta \sin 3\theta + \sin \theta \sin 5\theta + \sin \theta \sin 7\theta$$

同様に $(\cos(-x) = \cos x$ に注意)

$$\sin \theta \sin 3\theta = \frac{1}{2} (\cos 2\theta - \cos 4\theta)$$

$$\sin \theta \sin 5\theta = \frac{1}{2} (\cos 4\theta - \cos 6\theta)$$

$$\sin \theta \sin 7\theta = \frac{1}{2} (\cos 6\theta - \cos 8\theta)$$

辺々加えて

$$= \frac{1}{2} (\cos 2\theta - \cos 8\theta)$$

問題 6

次の式を 1 つの三角関数で表せ。

解答:

$$a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin(x + \varphi)$$

$$\text{ここで } \cos \varphi = \frac{a}{\sqrt{a^2 + b^2}}, \sin \varphi = \frac{b}{\sqrt{a^2 + b^2}}.$$

$$(1) -\sqrt{3} \sin x + \cos x$$

$$a = -\sqrt{3}, b = 1 \text{ だから } \sqrt{a^2 + b^2} = \sqrt{3+1} = 2$$

$$\cos \varphi = -\frac{\sqrt{3}}{2}, \sin \varphi = \frac{1}{2} \text{ より } \varphi = \frac{5}{6}\pi$$

$$\therefore -\sqrt{3} \sin x + \cos x = 2 \sin\left(x + \frac{5}{6}\pi\right)$$

$$(2) 3 \cos x + \sqrt{3} \sin x = \sqrt{3} \sin x + 3 \cos x$$

$$a = \sqrt{3}, b = 3 \text{ だから } \sqrt{a^2 + b^2} = \sqrt{3+9} = 2\sqrt{3}$$

$$\cos \varphi = \frac{\sqrt{3}}{2\sqrt{3}} = \frac{1}{2}, \sin \varphi = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2} \text{ より } \varphi = \frac{\pi}{3}$$

$$\therefore 3 \cos x + \sqrt{3} \sin x = 2\sqrt{3} \sin\left(x + \frac{\pi}{3}\right)$$

関数 $y = -\sin x + \cos x$ の最大値と最小値、およびそのときの x の値を求めよ。ただし $0 \leq x < 2\pi$ とする。

解答:

合成する. $a = -1, b = 1$ だから $\sqrt{a^2 + b^2} = \sqrt{2}$

$$\cos \varphi = -\frac{1}{\sqrt{2}}, \sin \varphi = \frac{1}{\sqrt{2}} \text{ より } \varphi = \frac{3}{4}\pi$$

$$y = \sqrt{2} \sin\left(x + \frac{3}{4}\pi\right)$$

$$0 \leq x < 2\pi \text{ より } \frac{3}{4}\pi \leq x + \frac{3}{4}\pi < \frac{11}{4}\pi$$

この範囲で $\sin$ が 1, -1 になる角を探す。

$$\text{最大: } \sin\left(x + \frac{3}{4}\pi\right) = 1 \text{ となるのは } x + \frac{3}{4}\pi = \frac{5}{2}\pi \left(\frac{\pi}{2}\right)$$

は範囲外)

$$x = \frac{5}{2}\pi - \frac{3}{4}\pi = \frac{7}{4}\pi$$

$$\text{最小: } \sin\left(x + \frac{3}{4}\pi\right) = -1 \text{ となるのは } x + \frac{3}{4}\pi = \frac{3}{2}\pi$$

$$x = \frac{3}{2}\pi - \frac{3}{4}\pi = \frac{3}{4}\pi$$

$$\therefore \text{最大値 } \sqrt{2} \left(x = \frac{7}{4}\pi\right), \text{ 最小値 } -\sqrt{2} \left(x = \frac{3}{4}\pi\right)$$